

The manuscript presents some machine learning and epidemic models to identify and prevent the spread of COVID-19.

In my opinion, the paper is quite good and deserves publication. However, there are some issues to fix. My remarks are listed as follows.

Abstract: It should be improved. The intentions of the authors are clear, but I think that the authors should strengthen the explanations of motivations and originality of their own proposal.

Introduction: It is quite good, but I think that the authors should add more comments about their motivation and contributions. They explain such factors, but some further sentences are needed.

Section 2:

- I like the way the authors express the mathematical contents. Just a unique remark: for Theorem 2.3, the proof is not rigorous. Can I suggest the authors revising such proof? For instance, who are the 3rd and 4th equations? I can imagine, but it is not clear.
- In Subsection 2.3, the authors state that system (2.1) is solved numerically through a Runge-Kutta method. I understand the intentions of the authors, but I kindly ask them to provide a more suitable explanation about numerical approaches for parameter estimations.
In my opinion, the authors should first describe the steps of resolution and then present the results. This could be the opportunity, for unexpert readers, to understand why the adoption of the Runge-Kutta method is necessary. Moreover, why is the presentation of the network structure introduced at page 11? I recommend careful revisions to explain the necessary steps for the numerical results.

Section 3: This section is quite good, but it could be suitable to move this section to another part of the paper. In my opinion, the authors should revise the structure of the manuscript by considering first theoretical notions and then the presentation of numerical results.

Section 4: I kindly ask the authors, if possible, to underline the possible differences among the various machine learning methods. They are clear to me, but I think that some further comments are needed.

References: They are quite good. However, I suggest the authors, if they want, to add the following works that deal with similar topics and/or similar numerical approaches:

<https://doi.org/10.3934/nhm.2024022>

<https://doi.org/10.3934/nhm.2024037>

<https://doi.org/10.1515/math-2024-0126>

In short, the paper is good, but some modifications are needed. At the actual state, my final decision is “major revisions”.